Fiscal Year:	FY 2011	**Task Last Updated:**	FY 03/07/2011
PI Name:	Khusid, Boris Ph.D.		
Project Title:	Novel Electro-hydrodynamic Technology for Gas-Liquid Phase Separation in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Zip Code:	07102-1982	**Congressional District:**	10
Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
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COI Name (Institution):	James, Conrad (Sandia National Laboratories)		
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Task Description:	Due to the lack of gravity, the removal of bubbles formed in a flowing fluid presents a challenge in space applications. The most straightforward application of the proposed field and flow driven bubble stripping and/or two phase flow management is in lab-on-a-chip (LOC) devices envisioned for space applications. Such devices inherently have air bubbles which are difficult to remove. Existing methods of the bubble removal in LOC systems appear to be of limited use for space applications due to different functional requirements. In the first year of our work on the project, we developed a novel method for the field and flow driven concentration and removal of air bubbles from a droplet of aqueous-based polymer solutions, such as saliva and urine, in a microgravity environment. The main idea behind our method is to apply an electric force directly to a droplet via capacitive coupling to external electrodes during the application of an alternating current (AC) voltage. To eliminate adverse effects of electro-chemical reactions at the fluid-electrode interface, a fluid droplet is placed on an electrically insulating film and serves as a floating electrode. Following the application of a sufficiently strong AC voltage, air bubbles accumulate on the droplet surface to form a		

Task Description:	layer of dense air froth possessing yield stress. The proposed method for the field and flow driven concentration and removal of air bubbles from droplets overcomes the critical challenges facing current technologies for gas-liquid separation and appears to meet the stringent requirements of LOC devices for testing bodily fluids in microgravity. During the second year, we developed a conveyer-belt technique that employs our method for continuous processing of many droplets. Experiments were conducted within the range of parameters appropriate for future applications of LOC devices in microgravity. As air bubbles are exposed to a buoyancy force in a normal gravity environment, we also conducted experiments on suspensions of neutrally buoyant particles dispersed in a saliva simulant to gain a better understanding of mechanisms underlying the field and flow driven segregation of bubbles in microgravity. The particle segregation is found to be a non-monotonic function of the field frequency. Increasing the voltage magnitude facilitates the particle segregation. A PhD student (minority) and a postdoc were involved in the research during the second year.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Due to the lack of gravity, the displacement and removal of bubbles formed in a flowing fluid presents a challenge in space applications. The inadvertent appearance and accumulation of vapor and gas bubbles in liquids can cause fluid systems, such as liquid pumps, volumetric techniques for sampling liquids, and biotechnology systems, fail to perform. The ongoing research effort should improve our understanding of the behavior of gas-liquid mixtures subject to a high-gradient electric field and facilitate the design and optimization of devices to mitigate these undesirable phenomena in microgravity. The research effort addresses the scientific and engineering issues prioritized as “Higher-Priority Areas” within the Fundamental Microgravity Research and Engineering Research Enabling Exploration programs by the Committee on Microgravity Research and as “Critical” and “Severely-Limiting” by the workshop on two-phase flow in microgravity. The method developed for stripping bubbles from a droplet offers the potential for an in-line debubbler in a pre-processing stage of LOC systems for bodily fluids in future space applications. In December 2010 we presented results of our study at the seminar in NASA GRC that was organized by Arnon Chait and Emily S. Nelson. It was decided to consider whether our method of the bubble removal can be used in open microchannels similar to that used by Nelson’s group in LOC systems for testing body fluids in microgravity. We are currently in the process of adjusting our method to the conditions of microfluidics provided by Nelson’s group
Task Progress:	The study was conducted within the scope of the originally-proposed research plan. Arnon Chait, Technical Officer, discussed opportunities at GRC for applications of the proposed electro-hydrodynamic technology for gas-liquid phase separation in microgravity. The consensus was that the most straightforward application of the proposed field and flow driven bubble stripping and/or two phase flow management is in lab-on-a-chip (LOC) devices now envisioned for space applications. Such devices inherently have air bubbles which are difficult to remove. Following this suggestion, we discussed with Emily S. Nelson, NASA GRC, properties of aqueous-based solutions and operating parameters of LOC devices envisioned for testing body fluids, such as saliva and urine, in microgravity. Existing methods of the bubble removal in LOC systems appear to be of limited use for space applications due to different functional requirements. Once the specimen of a bodily fluid has been collected, its composition has to be protected against chemical changes. Next, an LOC system should be able to process fluids having different physical properties. In the first year of our work on the project, we developed a new method for the field and flow driven concentration and removal of air bubbles from a droplet of aqueous-based solutions, such as saliva and urine, in a microgravity environment. The main idea behind our method is to apply an electric force directly to a droplet via capacitive coupling to external electrodes during an alternating current (AC) voltage signal. Following the application of an AC voltage, air bubbles accumulate on the droplet surface to form a layer of dense air froth possessing yield stress. Part of the droplet enriched with bubbles is then transferred to one of the external electrodes by applying a short voltage pulse. The proposed method overcomes the critical challenges facing current technologies for gas-liquid separation in microgravity and appears to meet the stringent requirements of lab-on-a-chip (LOC) devices for space applications. During the second year, we developed a conveyer-belt technique that employs the proposed method for continuous processing of many droplets. This technique consists of four key steps. To eliminate the adverse effects of electro-chemical reactions at the fluid-electrode interface, fluid droplets with air bubbles are situated on an electrically insulating film which is placed on a ground electrode mounted on a 3-D moving stage. As a droplet is positioned under the nozzle tip, a sufficiently strong AC voltage is applied to the nozzle. The applied AC voltage periodically stretches a droplet causing air bubbles to accumulate on the droplet surface and form a layer of dense froth. The application of a short voltage pulse and/or raising the stage forms a liquid bridge that allows us to pump the froth out. Under our experimental conditions, the relaxation time of charge within the fluid is short compared to the time scales of an applied field and the field induced fluid flow. As air bubbles are exposed to a buoyancy force in a normal gravity environment, we conducted experiments on the following suspensions of neutrally buoyant particles to confirm that our method for bubble removal does not require the presence of the buoyancy force exerted on air bubbles in ground-based experiments: • 0.5% -suspensions of hollow glass particles (Polysciences, size 2-20 μm, density 1.1 g/cm3) dispersed in low volatile polyethylene glycol (PEG 200, Sigma Aldrich, density 1.1 g/cm3, conductivity 3.0 μS/cm, viscosity 76 cP, surface tension 44 mN/m) • 1%-suspension of blue polyethylene microspheres (Cospheric, size 27-45 μm, density 1.0 g/cm3) dispersed in a saliva simulant [(w/w): DI-water, 0.65% sodium carboxymethyl cellulose, 1% glycerol, 0.05 % methyl para-hydroxybenzoate (all from Sigma Aldrich), density 1.01g/cm3, conductivity 1,100 μS/cm, viscosity 25 cP, surface tension 60 mN/cm]. It is found that particles segregate on the droplet surface under the energized nozzle and follow it as the nozzle moves. The particle segregation is found to be a non-monotonic function of the field frequency. Increasing the voltage magnitude is observed to facilitate the particle segregation. Planned work for the next year 1. We will continue the ground-based experiments to gain understanding of mechanisms underlying the field and flow induced gas-liquid separation.

	2. We will design, fabricate, and test several electro-hydrodynamic devices for the removal of air bubbles from fluids in open microchannels provided by Nelson's group, NASA GRC. 3. We will continue analytical and numerical analysis of models for the the field and flow driven gas-liquid separation. The theoretical predictions will be compared with results of ground-based experiments on air bubbles and neutrally buoyant particles.
Bibliography Type:	Description: (Last Updated: 02/06/2025)
Articles in Peer-reviewed Journals	Lo TS, Khusid B, Koplik J. "Field-induced alignment of flexible polyelectrolytes in solution." Physical Review Letters, 2010 May 28;104(21):218303. http://dx.doi.org/10.1103/PhysRevLett.104.218303 ; PMID: 20867144 , May-2010